

The Science of Memory Retention

A research notesheet, with implications for a retention-testing app

Why we forget

Ebbinghaus's forgetting curve (1885) still holds up under modern research: without review, memory decays fast and non-linearly. Most of what you lose, you lose in the first 24 to 48 hours, then the rate of loss slows. **Timing of review matters as much as amount of review.**

The two techniques with the strongest evidence

A widely cited 2013 review (Dunlosky, Rawson, Marsh, Nathan, and Willingham) rated ten popular study techniques against the research literature. Only two earned a “high utility” rating:

1. Practice testing (retrieval practice). Actively pulling information from memory beats restudying it, even with no feedback given. In Roediger and Karpicke's 2006 study, students who read a passage once and then repeatedly tried to recall it from memory outperformed students who reread it four times: a week later the retrieval group remembered substantially more. Testing isn't just a way to measure learning, it's a way to produce it. Retrieval works best when it requires actual production (short answer, free recall) rather than recognition (multiple choice).

2. Distributed practice (spacing). Spreading sessions out over time beats massing them together, regardless of material or task. A major meta-analysis (Cepeda et al., 2006) found the optimal gap between study sessions is roughly 10 to 20% of how long you need to remember the material. To hold something for a test in 30 days, review every 3 to 6 days, not the night before.

Moderate-utility techniques worth layering on top: interleaving (mixing topics instead of blocking by unit), elaborative interrogation (“why is this true?”), and self-explanation (explaining the reasoning behind a step or answer).

Low-utility techniques, worth deprioritizing as a primary mechanism: rereading, highlighting, summarizing, keyword mnemonics, imagery. They feel productive but the evidence for durable retention is weak.

Why it works: storage strength vs. retrieval strength

Robert Bjork's framework splits memory into *storage strength* (how entrenched a memory is) and *retrieval strength* (how easily it's accessible right now). Rereading boosts retrieval strength temporarily, it feels familiar, without doing much for storage strength. Retrieval practice does the opposite: the harder but still successful the retrieval, the more storage strength grows. Bjork calls conditions like spacing, interleaving, and testing “desirable difficulties.” They slow down apparent short-term performance but produce learning that lasts. Design for what a student remembers in a month, not what feels easy today.

The metacognition trap

Learners are bad judges of their own learning. Rereading and highlighting create a feeling of fluency (“this seems familiar, I must know it”), so students consistently rate these weaker strategies as effective and underuse testing and spacing, even after being told the research. This is sometimes called the fluency illusion. Any tool that lets learners freely choose how to study will tend to reproduce this bias unless it's deliberately designed around it.

Sleep and consolidation

Sleep, especially slow-wave sleep, actively consolidates new declarative memories. The hippocampus replays recently learned material, helping transfer it into more durable, distributed storage in the cortex. Material studied with a full night's sleep before the first retest tends to consolidate better than material crammed without sleep in between.

How apps operationalize this: SM-2 and FSRS

Flashcard apps formalize spacing with scheduling algorithms. SM-2 (1987, used by early Anki and SuperMemo) increases the interval before the next review each time a card is recalled correctly, and resets it on failure. FSRS, a newer algorithm, models three variables per item (difficulty, stability, predicted probability of recall), trained on hundreds of millions of real review logs, and schedules the next review right before predicted recall drops below a target retention rate (commonly ~90%). It's reported to need 20 to 30% fewer reviews than SM-2 for equivalent retention, though it needs meaningful review volume (roughly 1,000+ reviews) before it clearly outperforms SM-2.

Applying this to your product

1. **Make the core loop retrieval, not review.** Don't just resurface uploaded notes or slides for rereading. Auto-generate short-answer or free-recall questions from the material and default to production-based formats over multiple choice, since production testing produces a stronger effect than recognition testing.
2. **Build in a real spacing algorithm, not a fixed schedule.** SM-2 is simple to implement and good enough to start. FSRS is worth the added complexity once you have real review data, since it needs volume to outperform SM-2. Either is a reasonable build given your background.
3. **Delay the first test.** Testing immediately after exposure draws on short-term memory and doesn't build durable retention as well. A brief delay before the first retrieval attempt (hours, not seconds) produces better long-term results.
4. **Interleave, don't block.** If someone's studying multiple classes or chapters, mix questions across them instead of drilling one topic to exhaustion before moving on. It feels harder. That's the point.
5. **Fight the fluency illusion directly.** Before revealing the answer, ask users to predict their confidence. Comparing predicted versus actual performance over time makes the gap between “feels known” and “is known” visible, this could be a real differentiator versus a plain flashcard app.
6. **Design around sleep, not around cramming.** Where you control notification timing, favor a first review session after a night's sleep over same-day repetition.
7. **Reserve elaboration prompts for a tier above basic recall.** Once a user reliably recalls a fact, layer in “why” and “how does this connect to X” prompts for material where application matters, like problem sets, not just fact recall.
8. **Treat passive features (flip cards, highlight, reread notes) as secondary.** They're comforting and low-effort, but the research is clear they're the weakest lever available. If you include them, frame them as prep before active testing, not a substitute for it.

Key sources

Ebbinghaus (1885), Memory: A Contribution to Experimental Psychology · Cepeda, Pashler, Vul, Wixted, & Rohrer (2006), Psychological Bulletin · Roediger & Karpicke (2006), Psychological Science · Bjork & Bjork (2011, 2014), desirable difficulties · Dunlosky, Rawson, Marsh, Nathan, & Willingham (2013), Psychological Science in the Public Interest · Rasch & Born and related sleep-consolidation literature · Anki/FSRS documentation (Ye et al., open-spaced-repetition project)